

Modeling the Edge-Cloud Continuum: a Brain-Computer Interface Case Study

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Abstract—The availability of low-cost EEG headsets and advancements in signal processing have expanded the potential of Brain-Computer Interface (BCI) systems, particularly in cognitive buildings for enhanced accessibility and automation. However, a structured approach for integrating EEG-based applications within an edge-cloud continuum is lacking. This paper proposes a modeling approach for designing such applications and evaluates its effectiveness through a case study on steady-state visual evoked potential (SSVEP) recognition. Experiments conducted on four subjects show that SSVEP recognition is subject-dependent and influenced by electrode configuration, with classification accuracies ranging from 95.56% to 100% for individuals and 95.33% to 89.60% for aggregated data, with a Random Forest classifier. The proposed methodology lays the foundation for scalable, intelligent applications that leverage EEG signals to infer user preferences.

Index Terms—Brain-Computer Interface, Edge-Cloud Continuum, Modeling Approach, SSVEP, Wearables, Machine Learning, Deep Learning, Cognitive Buildings.

I. INTRODUCTION

The widespread availability of low-cost, wearable electroencephalogram (EEG) headsets [1] equipped with electrodes, combined with advancements in signal processing techniques and the increasing availability of computational power, has

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opened new possibilities for developing sophisticated Brain-Computer Interface (BCI) systems [1] [2] in cognitive environments [3]. These systems enable direct interaction between the human brain and external devices, eliminating the need for traditional input methods.

EEG signals can be recorded non-invasively from the scalp and translated into control signals, allowing users to communicate and interact with their surroundings without relying on peripheral nerves and muscles. This capability is particularly valuable in cognitive buildings, where understanding the needs of occupants is essential for enhancing both user experience and automation. The relevance of such technologies is even greater in buildings designed for individuals with impairments, where alternative interaction methods can significantly improve accessibility and autonomy. In this field, user-friendly EEG headsets [1] can serve as intuitive interfaces for interaction by recognizing specific EEG signals. These headsets can be integrated into an edge-cloud continuum architecture, enabling smart applications that efficiently leverage distributed computational resources across different layers of the continuum. In the literature, several studies have explored the use of EEG signals in smart applications [4]–[12]. Anyway, none of them has introduced a structured methodological approach for modeling such applications within the edge-cloud continuum.

To fill this gap, this paper introduces a modeling approach for designing systems deployed in the edge-cloud continuum. The proposed approach is particularly well-suited for applications that leverage wearable EEG sensors to infer user preferences within cognitive buildings. Additionally, the paper presents a case study demonstrating the proposed approach in practice, evaluating and comparing different algorithms for recognizing SSVEP [2] signals. Experiments were conducted on four subjects, and the results show that SSVEP signal recognition is subject-dependent and influenced by the number and configuration of electrodes other than by classification algorithms used.

The rest of the paper is organized as follows. Section II provides an overview of control signals used in EEG-based BCI systems, with a particular focus on the SSVEP paradigm and its application fields. Moreover, it will introduce some modeling approaches already presented in the literature. Section III describes the proposed modeling approach for developing applications within the edge-cloud continuum. The design and implementation of an SSVEP-BCI application, along with the experimental results obtained, is detailed in Section IV. Finally,

the study concludes with insights and suggestions for future research directions.

II. BACKGROUND AND STATE OF THE ART

The increasing availability of low-cost EEG headsets, advanced signal processing, and greater computational power enables the development of complex BCI systems for cognitive environments. These systems allow non-invasive EEG signal acquisition to translate human intentions into control commands, facilitating direct brain interaction with external devices.

There are several control signals detected by EEG headsets that can be used in Brain Computer Interface systems [2]. They can be divided into two main categories: *Event-Related* and *Spontaneous*. Among the Event-Related signals, the Event-Related Potentials (ERPs) refer to signals generated by the brain as a response to external sensory stimuli. By analogy with the response of a resonant circuit, the brain response to an external stimulus can be further divided into Transient Evoked Potential (TEP) and Steady State Evoked Potential (SSEP) responses. TEPs occur when the stimulation repetition frequency is less than 2 Hz, so that there is some resting period between two successive stimulations. If the repetition rate is higher than 6 Hz, a periodic response called the SSEP will result. P300 (a positive peak occurring nearly 300 ms after a rare or significant stimulus in an oddball paradigm) is considered transient ERP because it is a short-duration response to a specific stimulus rather than a sustained oscillatory pattern. On the contrary, SSEPs are reactions to a stimulus in which the constituent frequency components of the response remain constant with time in both amplitude and phase. Among SSEPs, Steady-State Visual Evoked Potentials (SSVEPs) and Steady-State Auditory Evoked Potentials (SSAEPs) are evoked by the occurrence of a visual and audio stimulus, respectively, and are often located in the occipital region. Unlike other systems, they are considered BCI-dependent because the production of the brain signal depends on the control of eye/ear movements through the outlets of the cranial nerves and ocular/ear muscles.

One of the most important fields in which SSVEP signals are used is on assisting disabled individuals in improving their quality of life through everyday tasks. One practical application target is the realization of a noninvasive, low-cost mind controlled speller aimed at improving the communication tasks of individuals suffering from severe neuromuscular diseases. The SSVEP-based BCI spellers proposed in [5], [13]–[15] consist of a visual flickering stimulation matrix resembling an alphanumeric keyboard or symbol control system used together with an EEG acquisition system and a classification processing unit for identifying the appropriate linguistic spelling target. Another use of the SSVEP-based BCI is aimed at establishing a direct connection between the brain and the wheelchair for improving the self-independence of disabled persons, like in [6] where a lightweight SSVEP-BCI system was embedded in a wheelchair for mind-control the movements. In [7], [16], an SSVEP-based lower limb exoskeleton for gait assistance was

proposed in order to help users wearing the exoskeleton to walk forward, turn right, turn left, sit, and stand by decoding electroencephalography signals in real-time.

In recent years, the exciting challenge of using the mind to control the external environment is motivating an ever-growing interest in using SSVEP-based BCI in different applications other than just medical and supporting disabled persons. Thus, the use of SSVEP-based BCI is being explored in fields like cognitive environments, gaming, and drone control. In [8], an SSVEP-based BCI was integrated into a smart home to control external devices like the entrance door, the wardrobe, the kitchens' worktop and drawers, and the light system. In [9], SSVEP-BCI replaced the player joystick in a hands-free video game. In [10], the authors proposed a visual interface for stimulating an SSVEP, allowing users to control (take-off, land, forward, and turn right) a drone in 3D physical space by using only EEG signals detected by a consumer-oriented EEG headset (Emotiv EPOC).

As seen above, several studies have explored the use of EEG signals for various purposes, but none of them have proposed a methodology for designing such systems.

Existing methodologies for modeling smart environments include the Smart Environment Metamodel (SEM) [17], which defines a unified vocabulary and metamodels for both functional and data-related aspects of SEs. Another approach to SE modeling is presented in [18], introducing a three-layer architecture based on the Model-Driven Architecture principles.

For Smart Objects (SO) and IoT systems, [19] proposes a metamodel framework that supports design, analysis, and implementation. Specifically, it defines three levels corresponding to the analysis, design, and implementation phases, ensuring seamless integration throughout the development process. Similarly, [20] introduces a three-layered architecture that provides suitable abstractions for developing IoT ecosystems with cognitive capabilities, leveraging computational resources at both the edge and the cloud.

Despite their effectiveness, these works reveal a gap in modeling abstractions for systems in the edge-cloud continuum. To address this, our paper introduces a novel methodology for their design.

III. THE PROPOSED MODELING APPROACH

This section introduces the modeling approach we specifically designed for the development of applications within the edge-cloud continuum. Compared to other approaches presented in the literature, our approach is, to the best of our knowledge, the first one to explicitly model systems distributed across the edge-cloud continuum. It achieves this by incorporating specific tags that identify the computational and storage resources required by each task within the system.

Figure 1 illustrates all the components that can be instantiated to model a specific edge-cloud system. More specifically, these components include: (i) *Sensors* and *Actuators*, represented as ellipses, which are responsible for acquiring data and sending commands, respectively; (ii) *Execution Tasks (ET)*,

depicted as rounded rectangles, which serve as the fundamental building blocks of the application or system. Each ET is associated with two circles that indicate its *Computational Complexity* and *Storage Complexity*, respectively.

The components can be connected by different types of arrows, each representing a specific kind of data flow. In particular, the dashed arrow indicates a continuous stream of data (*Data Flow*), typically generated by sensors that constantly sample specific physical quantities. The solid arrow represents the exchange of higher-level information between ETs (*Preprocessed Data Flow*) compared to the dashed arrow. This information may consist of filtered raw data, preprocessed data, or high-level knowledge. Finally, the dotted line indicates a potential path for transferring a trained model between two ETs (*Model Transfer*). This last type of connection enables pushing a model — possibly trained in the cloud — toward edge devices for deployment.

ETs can be positioned in different locations within the model based on their resource requirements. The greater the resource demand, the higher in the model the ETs are placed, representing devices with greater computational and storage capabilities.

Anyway, the position of ETs within the diagram does not strictly define where a specific task must be executed. During the modeling phase, the goal is to provide general guidance, taking into account factors such as the complexity of the task being integrated into the diagram. This complexity is indicated by the circled numbers in the bottom right corner of each task. Specifically, computational complexity is represented by a purple circle, while storage complexity is represented by a red circle. These values range from 1 to 4. Intuitively, a task with high computational and storage complexity is more likely to be executed in the cloud, although this does not exclude the possibility of it running at the edge when feasible. When considering task placement, it is also essential to evaluate the proximity of tasks to one another and to sensors or actuators. For instance, if a task Task_A is positioned close to a sensor Sensor_1, this may indicate that Task_A is designed to receive large volumes of data (in the form of Data Flow Messages) directly from Sensor_1, possibly via a dedicated connection. In such cases, Task_A will likely process the raw data by filtering it, generating aggregated insights, or extracting meaningful knowledge, which can then be transmitted to other tasks, such as Task_C and Task_D, using Processed Data Flow Messages. These tasks, i.e. Task_C and Task_D, may utilize the processed information for various purposes, such as assessing the quality of the received data, providing feedback, or training a model. The trained model can subsequently be transferred (through Model Transfer Messages) to another task, Task_N, capable of executing it. When required, this execution can trigger specific actuations within the target cognitive building, closing the loop between data collection, processing, and intelligent response.

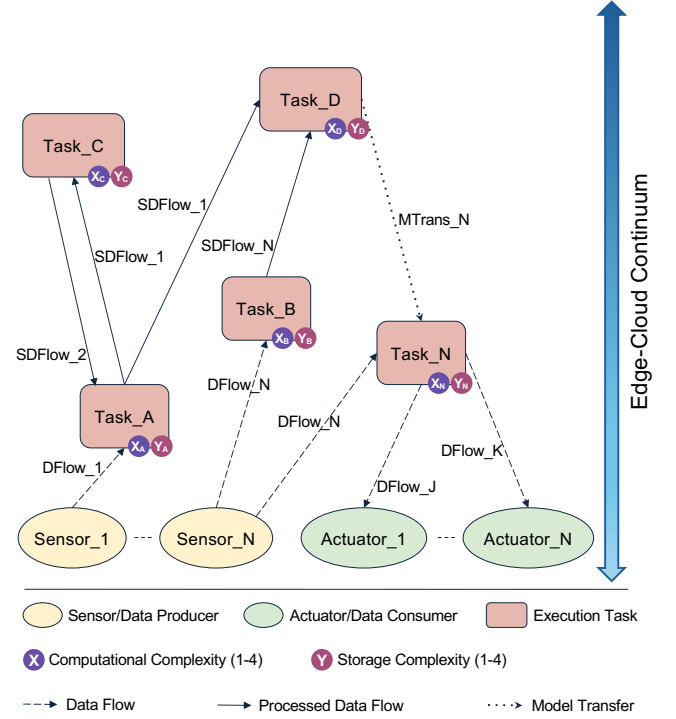


Fig. 1: The Proposed Modeling Approach.

IV. CASE STUDY

In order to demonstrate the proposed modeling approach, we exploited it for the development of a BCI application, as illustrated in Figure 2. More in detail, in the figure are shown (i) an *EEG_Diadem* Sensor block as the producer of EEG signals; (ii) a *Collection and Labeling* ET, which collects EEG data, filters and labels it; (iii) an *SSVEP_ModelMaker* ET, which trains a model to recognize SSVEP; (iv) a *SSVEP_Recognition* ET, which executes the trained model; and (v) a *Cognitive Building* Data Consumer block, which is the cognitive building receiving the recognized SSVEP inferred states.

The case study focuses on the accurate detection and classification of steady-state visual evoked potentials (SSVEPs) from EEG signals recorded using a wearable EEG headset, with the aim of implementing EEG-based BCI systems. Specifically, the experiment involved participants observing white squares flickering at different frequencies on a black background displayed on an LCD screen. During each trial, participants were instructed to focus either on one of the flickering squares or on the static black background (Figure 3).

EEG signals were recorded using the *DIADEM*¹ headset by BITBRAIN, a wearable dry-EEG device featuring 12 channels (Fp1, Fp2, AF7, AF8, F3, F4, P3, P4, PO7, PO8, O1, O2) [2] and a sampling rate of 256 Hz.

¹Bitbrain Diadem. <https://www.bitbrain.com/neurotechnology-products/dry-eeg/diadem>. Last seen 10/02/2025.

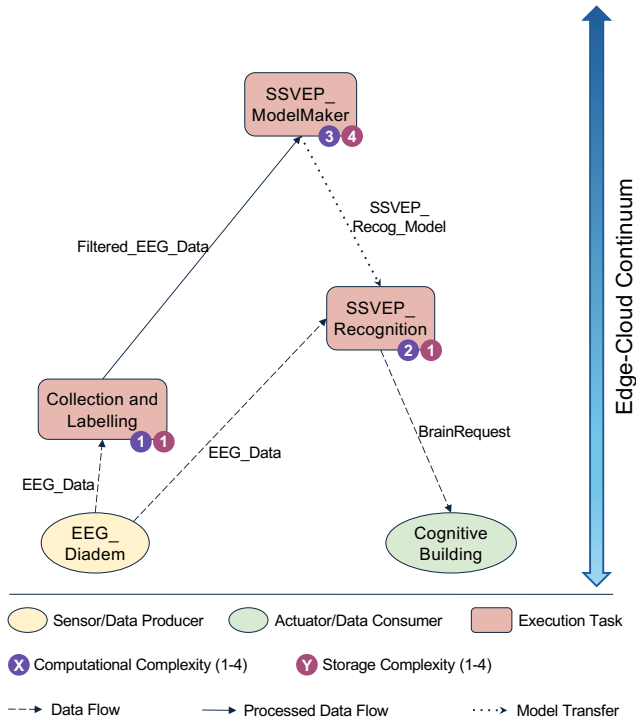


Fig. 2: The Designed BCI Application.

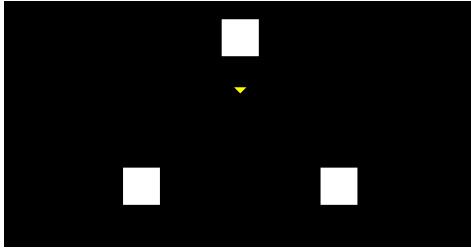


Fig. 3: SSVEP GUI

The experiment was conducted with four male participants aged 33 to 42 years. Each participant sat 65 cm away from the screen and performed the task twice. EEG signals were collected throughout the sessions for subsequent analysis.

The *Collection and Labelling* and *SSVEP_ModelMaker* ETs have already been implemented and validated and will be described below, while the *SSVEP_Recognition* ET, which enables real-time applications, is currently under development.

The Collection and Labelling Execution Task

The ET is responsible for generating visual stimuli and acquiring EEG signals from the headset (*EEG_Data*), selecting and filtering the relevant EEG channels (where SSVEP signals are observed), that are then processed by the *SSVEP_ModelMaker* ET (*Filtered_EEG_Data*).

Flickering frequencies for the squares were set at 8.57 Hz (bottom-right square), 10 Hz (top square), and 12 Hz (bottom-left square), selected to fall within the range most effective

for SSVEP stimulation [21] and, based on the 60 Hz refresh rate of the monitor, to correspond to 1/7, 1/6, and 1/5 of the refresh period, respectively. Additionally, a static black square in the center, matching the background color, was included. A directional arrow appeared above a target square or the central square, guiding participants on where to focus. During each trial, participants were instructed to focus on one square at a time in a randomized sequence, for a total of eight repetitions per square. The timing protocol for each trial was as follows:

- 1) An arrow appeared above the target square, indicating the square to focus on;
- 2) One second after the arrow's appearance, all squares began flickering at their designated frequencies for 7 seconds;
- 3) A 4-second rest period followed, during which no squares flickered.

Each trial lasted 12 seconds (i.e., 1 + 7 + 4 seconds), and the sequence was repeated 32 times (8 repetitions × 4 squares, including the static central square).

This ET was implemented on a computer running interconnected software applications that facilitated the presentation of the visual interface and concurrent EEG signals acquisition. Time-stamped markers were embedded in the EEG signals to indicate different phases of the experiment. The primary software tools used were (see Figure 4):

- **Bitbrain Device Viewer:** provided by the headset manufacturer, this software enables real-time visualization of EEG signals acquired via Bluetooth, along with the status of each electrode. It supports real-time export of EEG signals using the Lab Streaming Layer (LSL) protocol²;
- **OpenVibe³ Acquisition Server:** part of the OpenVibe suite, this server facilitates EEG signals acquisition through standard (e.g., LSL) or proprietary protocols for various EEG headsets. Since no specific driver was available for the DIADEM headset, the LSL protocol was used in conjunction with the Bitbrain Device Viewer to acquire EEG data;
- **OpenVibe Designer:** another component of the OpenVibe suite, this tool was used to develop the graphical interface displaying the flickering squares. It also handled the real-time reception of EEG signals from the Acquisition Server and saved the data as CSV files for subsequent analysis.

This integrated setup ensured seamless coordination between visual stimulus presentation and EEG data collection, with precise time markers enabling robust experimental analysis.

The SSVEP_ModelMaker Execution Task

This ET is responsible for analyzing the signals acquired during the experiments, covering the entire process from

²Lab Streaming Layer (LSL). <https://labstreaminglayer.org/>. Last seen 10/02/2025.

³OpenVibe - Software for Brain-Computer Interface and Real Time Neurosciences. <https://openvibe.inria.fr/>. Last seen 10/02/2025.

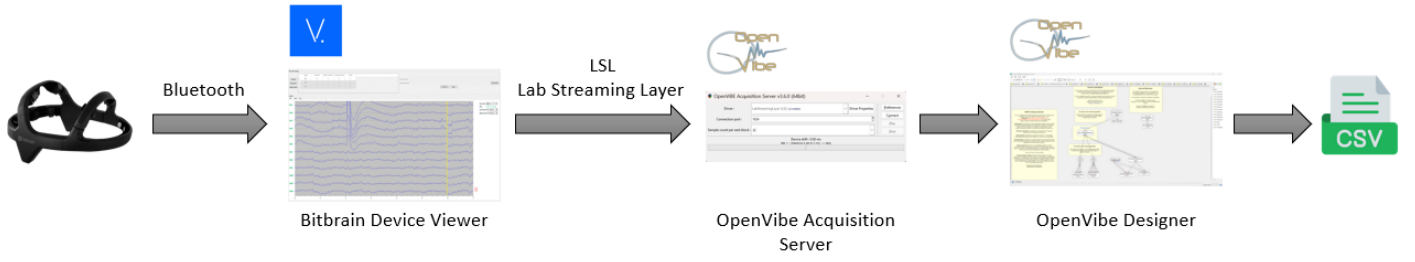


Fig. 4: Collection and Labelling ET Operations

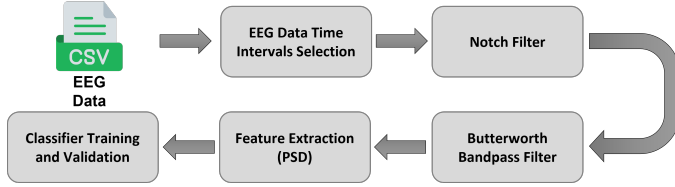


Fig. 5: SSVEP_ModelMaker Task Operations

preprocessing to the implementation and validation of machine learning (ML) and deep learning (DL) models for classifying the SSVEP signals. This process is carried out through Python scripts and is shown in Figure 5.

The preprocessing begins by selecting the time intervals of interest in *Filtered_EEG_Data*, corresponding to the observation of the flickering squares. These signals are then processed using a 50 Hz Notch filter to remove power line interference and a Butterworth bandpass filter between 6 and 30 Hz to preserve the fundamental frequency and the first harmonic for each flickering frequency [22]. For each obtained filtered 7-second interval and for each individual EEG channel, the Power Spectral Density (PSD) is calculated, by applying the Welch method [23], and then used as a feature for the classification algorithms.

Several classification algorithms have been tested, including LDA, SVM, KNN, Decision Tree, Random Forest, and 1D-CNN. To validate the classifiers, a 5-fold cross-validation method has been used. The results obtained for the different algorithms and various subsets of EEG channels are presented. Furthermore, data aggregation is examined for both the combined dataset - with classifiers trained and tested on data from all subjects, as detailed in Table I and Figure 6 - and individual subjects - with classifiers trained and tested on data from the same subject, as illustrated in Figures 7-10.

Table I shows that Random Forest consistently achieved the highest classification accuracy across all electrode configurations, with accuracies ranging from 89.60% (O2) to 95.33% (PO7, O1, O2, PO8). SVM and CNN also performed well, particularly with more electrodes, achieving 93.25% and 93.77%, respectively, in the (PO7, O1, O2, PO8) configuration. However, they showed a slight drop in performance when fewer electrodes were used, indicating that they are well-suited for identifying frequency-specific features in SSVEP

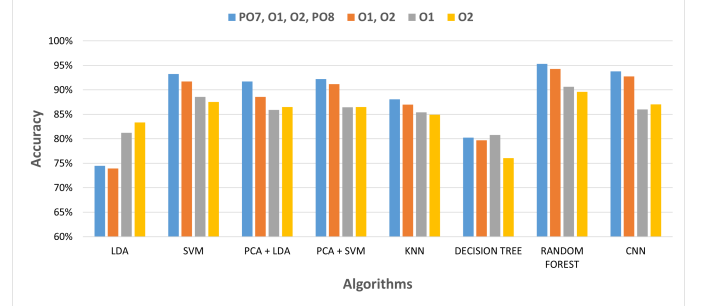


Fig. 6: Classification Accuracy per Algorithm and Electrode Configuration for all the subjects

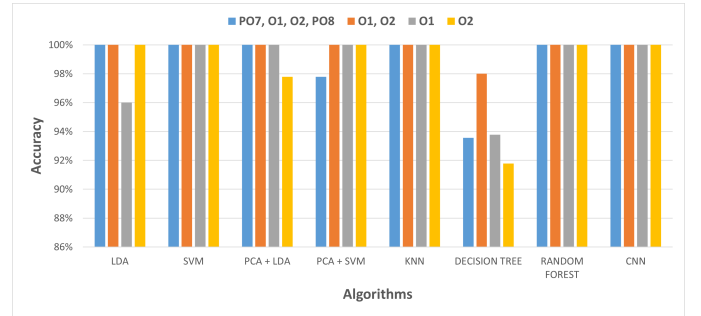


Fig. 7: Classification Accuracy per Algorithm and Electrode Configuration - Subject S1

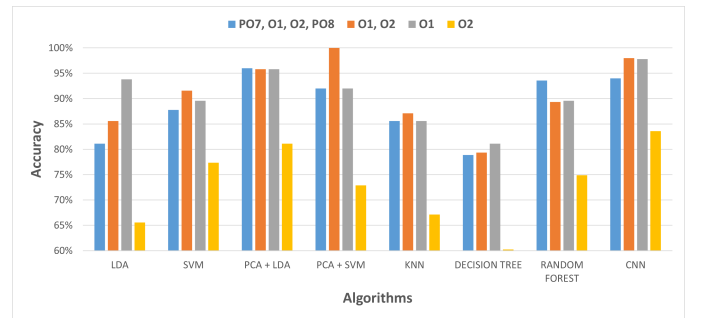


Fig. 8: Classification Accuracy per Algorithm and Electrode Configuration - Subject S2

Algorithm	PO7, O1, O2, PO8	O1, O2	O1	O2
LDA	74.47%	73.95%	81.24%	83.36%
SVM	93.25%	91.70%	88.56%	87.54%
PCA + LDA	91.70%	88.56%	85.92%	86.49%
PCA + SVM	92.21%	91.17%	86.45%	86.50%
KNN	88.06%	86.99%	85.43%	84.91%
Decision Tree	80.26%	79.70%	80.77%	76.06%
Random Forest	95.33%	94.28%	90.65%	89.60%
CNN	93.77%	92.73%	85.99%	87.02%

TABLE I: Classification Accuracy by Algorithm and Electrode Configuration for all the subjects.

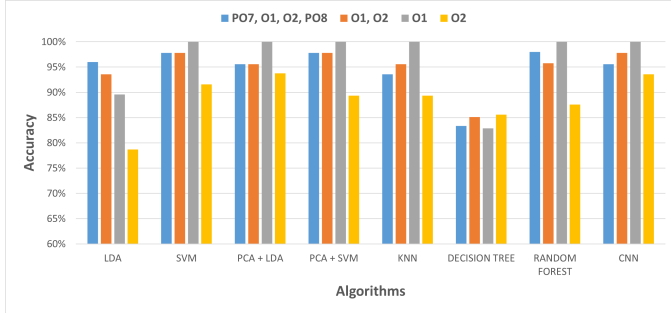


Fig. 9: Classification Accuracy per Algorithm and Electrode Configuration - Subject S3

signals, especially when spatial information from multiple electrodes is available. In contrast, LDA, KNN and Decision Tree algorithms showed lower performance. The use of PCA combined with classifiers (PCA + LDA and PCA + SVM) showed consistent performance improvements over standalone LDA, particularly in the (PO7, O1, O2, PO8) and (O1, O2) configurations, while SVM doesn't show same benefits.

The results highlight the importance of electrode selection on classification accuracy. Configurations with more electrodes (e.g., PO7, O1, O2, PO8) consistently yielded higher accuracies compared to configurations with fewer electrodes. For instance, CNN achieved 93.77% accuracy with the full set of electrodes but only 85.99% and 87.02% for O1 and O2, respectively.

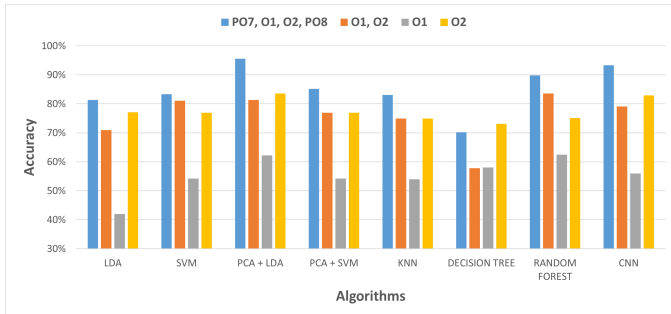


Fig. 10: Classification Accuracy per Algorithm and Electrode Configuration - Subject S4

Interestingly, even with fewer electrodes, some algorithms maintained relatively high accuracy. For example, Random Forest achieved 90.65% accuracy using only the O1 electrode and 89.60% with O2 alone. This highlights the potential for optimizing electrode configurations to balance classification performance and hardware constraints, for example in portable BCIs, characterized by reduced computational resources and wearable EEG systems with limited channels. On the other hand, systems prioritizing maximum classification performance may benefit from using configurations with broader spatial coverage, such as PO7, O1, O2, and PO8.

Regarding single subject results, shown in Figures 7-10, significant inter-subject variability is observed. Subject S1 consistently achieves near-perfect accuracies across all algorithms and channel subsets, indicating strong and reliable SSVEP responses. Conversely, Subject S4 exhibits markedly lower performance, with accuracies below 60% for certain algorithms and channel subsets. Algorithm robustness varies significantly. LDA and SVM (with PCA) consistently deliver high accuracy and reliability (comparable to Random Forest and CNN), making them strong candidates for real-time BCI systems. In contrast, KNN and Decision Tree exhibit greater variability and lower performance, especially for S4.

Also channel selection plays a crucial role in performance. The broader (PO7, O1, O2, PO8) subset often achieves the highest accuracies across subjects, while narrower subsets (e.g., O1 or O2) often show reduced performance, particularly for challenging cases like S4. This suggests that wider channel coverage may improve robustness but requires trade-offs in wearability for practical applications. In addition, significant differences in performance between O1 and O2 channels across different subjects are observed. So, it is suggested to use at least (O1, O2) configuration rather than single channel configurations.

V. CONCLUSIONS

This paper proposes a novel modeling approach for designing EEG-based BCI applications within an edge-cloud framework, enabling scalable and intelligent systems that efficiently leverage distributed computational resources. The proposal provides a structured approach to modeling application components, ensuring adaptability to real-time constraints and hardware-limited environments. The approach is demonstrated and validated through a case study on steady-state visual evoked potential (SSVEP) recognition, in which we have shown how Execution Tasks have been implemented and which type of data are transferred between them. The results confirm the feasibility of the proposed approach, showing high classification performance while highlighting the subject-dependent nature of SSVEP recognition, which can be addressed by optimization of electrode configurations and classification algorithms. More in detail, for individual subjects, the best classification accuracies achieved ranged from 95.56% to 100%, with significant variations in the predominance of

specific electrodes in SSVEP signal generation. LDA and SVM (with PCA) consistently achieved high accuracy and reliability, comparable to Random Forest and CNN. Additionally, aggregating data from all subjects yielded highly positive results, with classification accuracy ranging from 95.33% using the (PO7, O1, O2, PO8) configuration to 89.60% using the O2 electrode configuration, with a Random Forest classifier.

In conclusion, by leveraging the edge-cloud continuum, the proposed methodology enables scalable, intelligent and real-time EEG-based BCI applications, such as those in cognitive buildings, that efficiently infer user preferences through non-invasive methods.

Future research will focus on refining and extending the methodology to broader EEG-based applications, as well as completing the development of the proposed custom model with real-time functionality. Moreover, we plan to extend the experimental validation by involving a significantly more extensive and diverse set of subjects, aiming to mitigate the subject-dependent variability observed in EEG-based applications. Additionally, future efforts will explore potential challenges in real-world implementations, including assessing the impact of hardware constraints such as limited computational power and memory availability at the edge nodes.

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